

Adaptive Doppler Wrapping Method for Crab Angle Compensation in Airborne Early Warning Radar

Lang Xia

School of Integrated Circuits (School of
Information Science and Electronic
Engineering)
Shanghai Jiao Tong University
Shanghai, China
xl2023@sjtu.edu.cn

Muyang Zhan

Shanghai Aerospace Electronics
Technology Institute
Shanghai, China
zhanmuyang1992@126.com

Lingyu Wang

School of Integrated Circuits (School of
Information Science and Electronic
Engineering)
Shanghai Jiao Tong University
Shanghai, China
wly123@sjtu.edu.cn

Zihao Zou

Aerospace Information Research
Institute, Chinese Academy of Sciences
Beijing, China
zouzihao@aircas.ac.cn

Jingtao Ma

the College of Information, Mechanical
and Electrical Engineering
Shanghai Normal University
Shanghai, China
majingtao@shnu.edu.cn

Qing Ling

the College of Electronic Engineering
Naval University of Engineering
Wuhan, China
ling_qing1997@163.com

Jiaqi Wang

School of Integrated Circuits
(School of Information Science and
Electronic Engineering)
Shanghai Jiao Tong University
Shanghai, China
wjq9785@sjtu.edu.cn

Yuxing Zhao

School of Integrated Circuits
(School of Information Science and
Electronic Engineering)
Shanghai Jiao Tong University
Shanghai, China
zyx_0303@sjtu.edu.cn

Penghui Huang

School of Integrated Circuits
(School of Information Science and
Electronic Engineering)
Shanghai Jiao Tong University
Shanghai, China
huangpenghui@sjtu.edu.cn

Abstract—To address the range-dependent nonstationary clutter caused by the crab angle in airborne early warning (AEW) radars, this paper proposes a prior-free adaptive Doppler warping (ADW) compensation method. In this method, the Doppler centroids of the clutter are initially estimated for all range cells using fast Fourier transform (FFT). Then, a negative even-order power series model is fitted to these centroid estimates against range via the least-squares method to reject outliers. Finally, a compensation matrix is constructed from the fitted centroid curve, thus effectively compensating for the clutter nonstationarity. Simulation results demonstrate a significant enhancement in clutter suppression performance.

Keywords—airborne early warning radar, adaptive Doppler warping, crab angle, nonstationary clutter

I. INTRODUCTION

The high velocity of the airborne early warning (AEW) radar platform induces a broadening of the clutter spectrum, which necessitates effective clutter suppression to ensure reliable target detection and tracking. Conventional one-dimensional filters in either the temporal or spatial domain, due to their limited degrees of freedom (DOF), are often insufficient for canceling strong clutter with inherent space-time coupling. To address this, space-time adaptive processing (STAP) employs two-dimensional joint filtering in the spatial and temporal domains [1], [2], [3]. This technique is capable of suppressing mainlobe clutter while

simultaneously achieving coherent integration of target energy, thereby significantly improving the output signal-to-clutter-plus-noise ratio (SCNR).

However, the performance of STAP critically depends on the accurate estimation of the clutter covariance matrix. Under an ideal side-looking uniform linear array (ULA) configuration, the clutter exhibits homogeneous statistical properties across different range cells [4], [5], [6]. This permits the estimation of the covariance matrix for the cell under test (CUT) using training samples from adjacent range cells. In practice, however, factors such as crab angle induced by atmospheric disturbances disrupt this consistency, causing the space-time characteristics of clutter to vary across range cells [7], [8]. Consequently, the estimated covariance matrix for the CUT becomes mismatched, leading to a widened notch in the adaptive filter and degradation of STAP performance.

To compensate for non-stationary clutter, several methods have been developed for various array configurations. Predominant approaches include Doppler warping (DW) [9], angle-Doppler compensation (ADC) [10], and adaptive angle-Doppler compensation (A²DC) [11], [12]. Their principle is to align clutter data from all range cells to a reference Doppler or angle-Doppler cell by estimating and compensating for the range-dependent shift of the clutter spectrum centroid, thereby addressing the non-stationarity of mainlobe clutter. The space-time interpolation (STINT) method, which employs a predetermined transformation matrix to project clutter from all range gates onto a reference clutter subspace for

This work was supported in part by the National Natural Science Foundation Program of China under Grants 62501401, supported by Shanghai QiYuan Innovation Foundation, in part by the USCAST2023-29 and SAST2024-026. (Corresponding Author: Penghui Huang).

compensation [13], [14]. The derivative-based updating (DBU) also mitigate range dependence by assuming a linear relationship between range gates and the STAP weight vector [15]. However, the practical application of these existing methods is constrained by their requirement for prior knowledge of the space-time relationship, the use of insufficiently accurate fitting models, as well as high computational costs.

Inspired by the prior works, this paper proposes an adaptive Doppler warping (ADW) compensation method to address the non-stationary clutter problem in AEW radar with crab angle. The proposed method first employs a fast Fourier transform (FFT) to obtain a preliminary estimate of the clutter Doppler centroid for each range cell. Subsequently, it fits these estimated centroids using a range-dependent model based on a series of negative even powers to smooth the clutter Doppler centroid frequency. Finally, a compensation matrix is constructed using the fitted centroid values to accomplish clutter compensation. Compared with conventional adaptive approaches [10], [11], [12], the proposed method offers a more accurate fitting model. Furthermore, it achieves lower computational cost than the methods in [13], [14], [15].

The remainder of this paper is organized as follows. Section II presents the signal model and problem formulation. Section III elaborates on the proposed methodology in detail. Section IV provides the simulation experiments and performance analysis. Finally, Section V concludes the paper.

II. SIGNAL MODEL AND PROBLEM DESCRIPTION

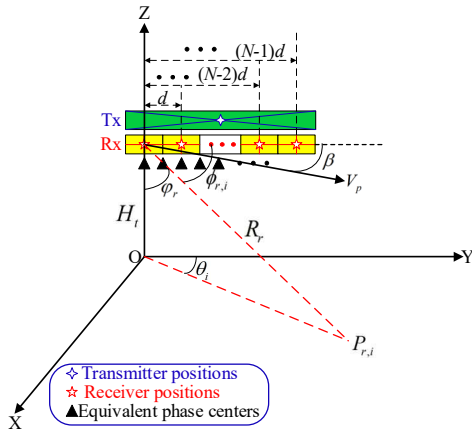


Fig. 1. Three-dimensional geometry between the airborne radar platform and a ground static clutter patch with the consideration of crab angle.

Fig. 1 presents the observation geometry within an $O\text{-}XYZ$ Cartesian coordinate system for a side-looking airborne radar incorporating a crab angle, β . In this system, the X -axis is defined as the range direction, the Y -axis aligns with the along-track (azimuth) direction of the antenna plane, and the Z -axis completes the right-handed orthogonal frame. The radar signal is transmitted by the full antenna aperture and received through N channels uniformly spaced by a distance d . The initial coordinates of the n th equivalent phase center, where $n = 1, 2, \dots, N$, are given by $(0, (n-1)d/2, H_t)$, with H_t and V_p representing the platform altitude and velocity, respectively. For the i th azimuth clutter patch in the r th range cell, let its elevation, azimuth, and cone angles be denoted by ϕ_r , θ_i , and $\phi_{r,i}$, respectively, where $\cos \phi_{r,i} = \sin \phi_r \cos \theta_i$.

As shown geometrically in Fig. 1, the spatial frequency $f_{s,r,i}$ and the normalized Doppler frequency $\bar{f}_{d,r,i}$ of the i th clutter patch in the r th range ring are respectively denoted as:

$$\begin{cases} f_{s,r,i} = \frac{d}{\lambda} \sin \varphi_r \cos \theta_i, \\ \bar{f}_{d,r,i} = \frac{2V_p}{\lambda F_{PRF}} \sin \varphi_r \cos(\theta_i - \beta) \end{cases} \quad (1)$$

where λ and F_{PRF} denote the wavelength of the electromagnetic wave and pulse repetition frequency, respectively.

Based on (1), the coupled expression for the normalized Doppler frequency $\bar{f}_{d,r,i}$ and the spatial frequency $f_{s,r,i}$ can be derived as:

$$\left(f_{s,r,i} - \xi \cos \beta \bar{f}_{d,r,i}\right)^2 + \left(\xi^2 \bar{f}_{d,r,i}^2 - \Omega_r^2\right) \sin^2 \beta = 0 \quad (2)$$

where $\xi = \frac{dF_{PRF}}{2V_p}$ and $\Omega_r = \frac{d}{\lambda} \sin \varphi_r$.

According to (2), the presence of a crab angle results in a range-dependent non-zero term $(\xi^2 \bar{f}_{d,r,i}^2 - \Omega_r^2) \sin^2 \beta$, leading to misaligned clutter ridges across different range rings in the space-time plane. This indicates that the clutter exhibits range dependence, which severely compromises the independent and identically distributed (i.d.d.) property of the range training samples. To address this issue, a non-stationary clutter compensation method is proposed in the following.

III. PROPOSED METHOD

From (1), the clutter Doppler centroid $f_{d,r,center}$ in the r th range gate can be rewritten as:

$$\begin{aligned} f_{d,r,center} &= \frac{2V_p}{\lambda} \sin \varphi_r \cos(\theta_{center} - \beta) \\ &= \frac{2V_p}{\lambda} \cos(\theta_{center} - \beta) \sqrt{1 - \frac{H_t^2}{R_r^2}} \end{aligned} \quad (3)$$

where θ_{center} is the azimuth angle of beam center.

Using a second-order Taylor expansion, (3) can be further approximated as:

$$f_{d,r,center} \approx \frac{2V_p}{\lambda} \cos(\theta_{center} - \beta) \left(1 - \frac{H_t^2}{2R_r^2} - \frac{H_t^4}{8R_r^4} \right) \quad (4)$$

$$= c_1 + c_2 R_r^{-2} + c_3 R_r^{-4}$$

where

$$\begin{cases} c_1 = \frac{2V_p}{\lambda} \cos(\theta_{center} - \beta) \\ c_2 = -\frac{V_p H_t^2}{\lambda} \cos(\theta_{center} - \beta) \\ c_3 = -\frac{V_p H_t^2}{4\lambda} \cos(\theta_{center} - \beta) \end{cases} \quad (5)$$

As seen from (4), the clutter Doppler centroid can be modeled as a series in terms of negative even powers of range. Therefore, after the clutter Doppler centroids for different range rings are preliminarily estimated using spectral analysis techniques such as the fast Fourier transform (FFT), the following expression can be obtained:

$$\mathbf{G}\mathbf{c} = \mathbf{f}_{d,center} \quad (6)$$

where $\mathbf{f}_{d,center} = [\mathbf{f}_{d,1,center}, \mathbf{f}_{d,2,center}, \dots, \mathbf{f}_{d,L,center}]^T$ is defined as the vector of clutter Doppler centroids preliminarily estimated using the FFT. L is the number of range gates. $\mathbf{c} = [c_1, c_2, c_3]^T$ is the vector of coefficients to be fitted. Matrix $\mathbf{G}$ can be expressed as:

$$\mathbf{G} = \begin{bmatrix} 1 & R_1^{-2} & R_1^{-4} \\ 1 & R_2^{-2} & R_2^{-4} \\ \vdots & \vdots & \vdots \\ 1 & R_L^{-2} & R_L^{-4} \end{bmatrix} \quad (7)$$

Hence, using the least squares method, the estimate of $\mathbf{c}$ is given by:

$$\hat{\mathbf{c}} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{f}_{d,center} \quad (8)$$

where the superscript T denotes the transpose.

Consequently, the clutter Doppler centroids for different range gates can be fitted as follows:

$$\hat{\mathbf{f}}_{d,r,center} = \hat{c}_1 + \hat{c}_2 R_r^{-2} + \hat{c}_3 R_r^{-4} \quad (9)$$

Then, we construct the Doppler frequency compensation matrix for the r th range gate as follows:

$$\mathbf{T}_{t,r} = \text{diag}\{\exp[j2\pi \mathbf{f}_{d,r,com} / F_{PRF} \mathbf{t}]\} \quad (10)$$

where the notation $\text{diag}\{\cdot\}$ converts a vector into a diagonal matrix. $\mathbf{f}_{d,r,com} = \hat{\mathbf{f}}_{d,r,center} - \hat{\mathbf{f}}_{d,ref,center}$ represents the Doppler frequency offset to be compensated in the r th range gate and $\hat{\mathbf{f}}_{d,ref,center}$ is the clutter Doppler centroid of the reference range gate. K is the number of pulses.

Finally, clutter compensation for the snapshot data at the r th range gate is performed as follows:

$$\mathbf{x}_{ref} = \mathbf{T}_r^H \mathbf{x}_r \quad (11)$$

where $\mathbf{x}_{ref} \in \mathbb{C}^{NK \times 1}$ and $\mathbf{x}_r \in \mathbb{C}^{NK \times 1}$ represent the space-time data of the reference range gate and the range gate to be compensated, respectively. $\mathbf{T}_r = \mathbf{T}_{t,r} \otimes \mathbf{I}_{N \times N}$ is the space-time compensation matrix and $\mathbf{I}_{N \times N}$ is the identity matrix. $\otimes$ is the Kronecker product operator. The superscript H denotes the conjugate transpose.

IV. SIMULATION ANALYSIS

In this section, based on the AEW radar simulation parameters provided in Table I, simulation verification of the proposed ADW method is conducted.

TABLE I SIMULATION PARAMETERS

Parameter	Value
Platform height	8000 m
Platform velocity	150 m/s
Carrier frequency	1.2 GHz
Pulse repetition frequency	2400 Hz
Pulse number	72
Spatial channel number	8
Physical channel spacing	0.125 m
Azimuth antenna size	1 m
Elevation antenna size	0.5 m

Fig. 2(a) shows the simulated airborne clutter Doppler spectrum under a crab angle of 30° . It can be observed that due to the platform motion, the clutter spectrum undergoes

broadening, resulting in the obscuration of the target signal. The crab angle also introduces significant range dependence in the near-range clutter. Correspondingly, the clutter space-time spectrum in Fig. 2(b) exhibits a semi-elliptical trajectory. This non-zero crab angle further causes noticeable broadening and dispersion of the clutter energy in the space-time plane.

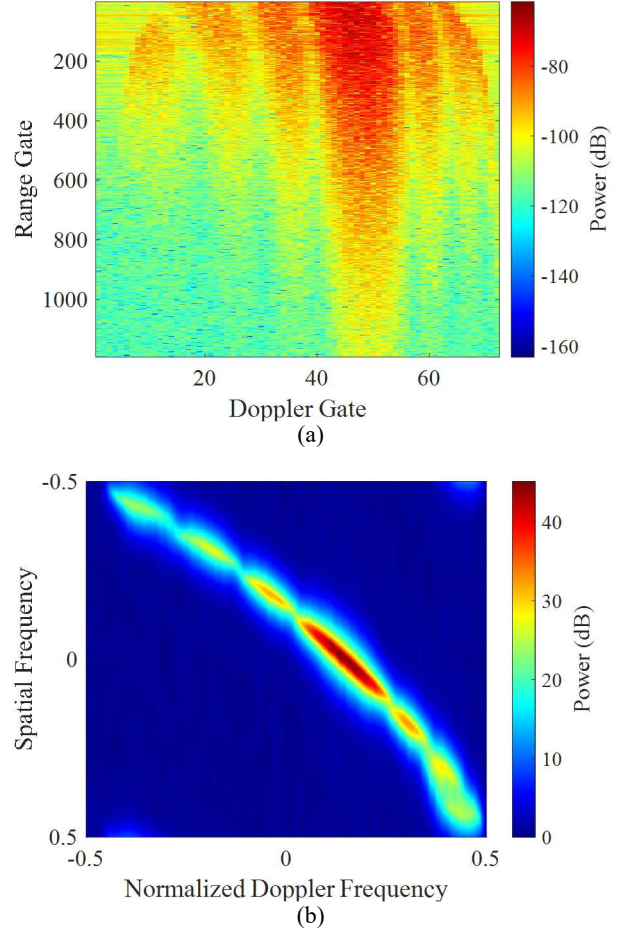


Fig. 2. Simulated airborne clutter. (a) Range-Doppler spectrum. (b) Space-Time spectrum.

Fig. 3(a) shows the range-varying curve of the clutter spectrum center frequency estimated via FFT. Due to noise and internal clutter fluctuations, the direct estimation results in severe frequency fluctuations, obscuring a clear range-dependent trend. In contrast, Fig. 3(b) displays the curve fitted with a range-based negative even-power series model, which reveals that the non-stationarity of the airborne clutter is relatively weaker at far range. Fig. 3(c) shows the range-Doppler spectrum after applying the proposed compensation method, where the range dependence of the clutter is significantly suppressed. Fig. 3(d) presents the compensated space-time spectrum, demonstrating effective focusing of clutter energy and a more concentrated distribution in the space-time plane. Fig. 3(e) presents the clutter suppression result obtained by applying the Extended Factored Approach (EFA) after compensation, where the clutter mainlobe is effectively suppressed, revealing the target signal clearly [16]. Fig. 3(f) compares the SCNR performance degradation after clutter suppression with and without the proposed compensation. The proposed method achieves a significant reduction in this loss, thereby validating its effectiveness.

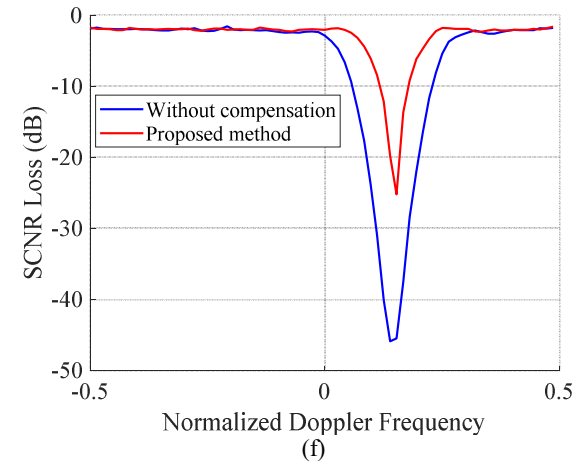
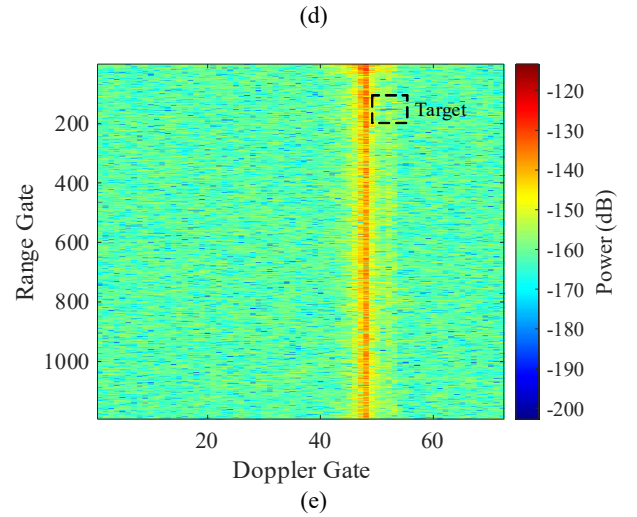
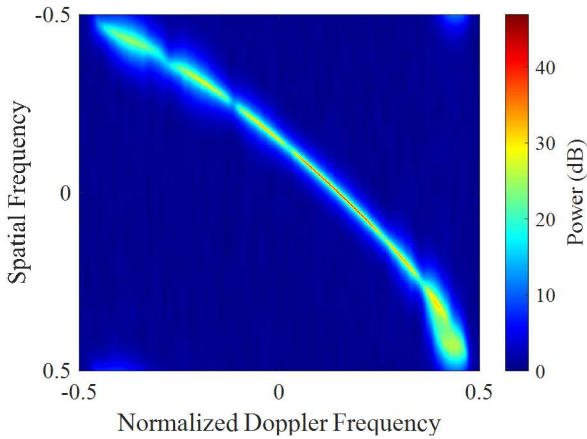
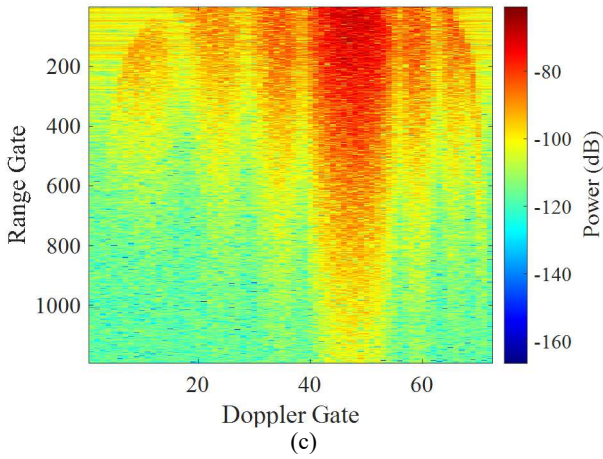
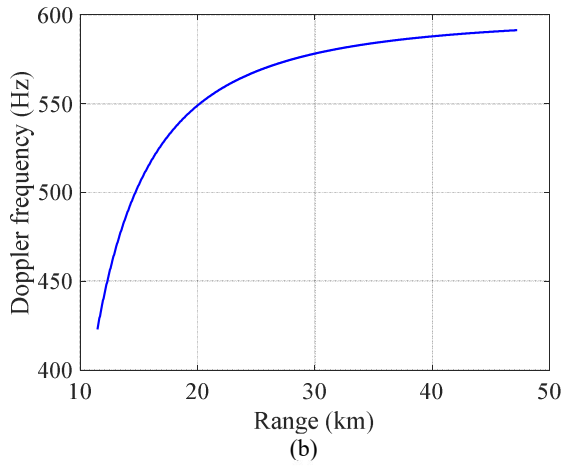
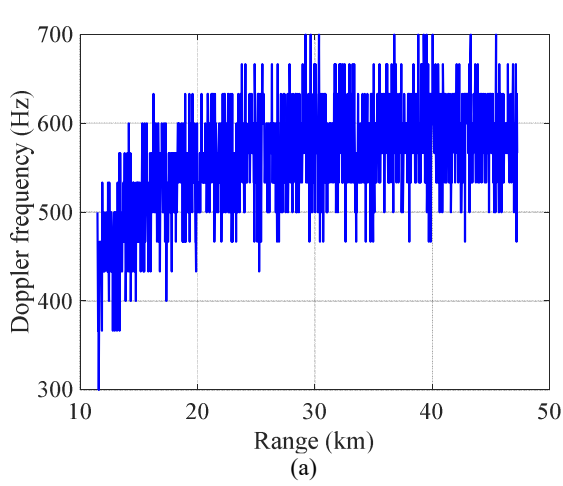


Fig. 3. Processing results of the proposed method. (a) Clutter Doppler centroid estimates based on FFT. (b) Corresponding model fitting results. (c) Compensated range-Doppler spectrum. (d) Compensated space-time spectrum. (e) Clutter suppression results based on the compensated data. (f) Comparison of the output SCNR loss curves (with and without compensation).

V. CONCLUSION

This paper tackles the clutter range dependence (non-stationarity) in AEW radar with a crab angle by proposing an adaptive Doppler warping compensation method. The key contributions are: 1) deriving and validating a negative even-power series model for the clutter Doppler centroid versus range; 2) developing a FFT-based estimator for the clutter spectrum center that requires no prior knowledge of platform motion. Simulations confirm the effectiveness of the proposed method in mitigating range dependence and improving clutter suppression.

REFERENCES

- [1] L. E. Brennan and L. S. Reed, "Theory of adaptive radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-9, no. 2, pp. 237–252, Mar. 1973.
- [2] Y.-L. Wang and Y.-N. Peng, *Space-Time Adaptive Processing*. Beijing, China: Tsinghua University Press, 2000.
- [3] J. R. Guerci, *Space-Time Adaptive Processing for Radar*. Boston, MA, USA: Artech House, 2003.
- [4] Y. Li, K. Duan, Z. Qiu, and Y. Wang, "Reduced-dimension STAP method via beamforming joint subarray synthesis for space-based early warning radar," *IEEE Sensors J.*, vol. 24, no. 22, pp. 37404–37419, 15 Nov. 2024.

- [5] Y. Wu, T. Wang, J. Wu, and J. Duan, "Training sample selection for space-time adaptive processing in heterogeneous environments," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 691–695, Apr. 2015.
- [6] Y. Xiong, W. Xie, Y. Wang, W. Chen, and M. Hou, "Short-range nonstationary clutter suppression for airborne KA-STAP radar in complex terrain environment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 2766–2776, 2025.
- [7] J. Chen, B. Liang, J. Zhang, D. Yang, Y. Deng, and M. Xing, "Efficiency and robustness improvement of airborne SAR motion compensation with high resolution and wide swath," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [8] Y.-L. Shi, "Impact of spatially and temporally nonstationary clutter on covariance matrix estimation," in *IET International Radar Conference 2013*, Xi'an, China, 2013, pp. 1–4.
- [9] C. Heng, B. Lim, E. Aboutanios, and B. Mulgrew, "Modified JDL with Doppler compensation for airborne bistatic radar," in *Proc. IEEE Int. Radar Conf.*, Arlington, VA, USA, May 2005, pp. 854–858.
- [10] B. Himed, Y. Zhang, and A. Hajjari, "STAP with angle-Doppler compensation for bistatic airborne radars," in *Proc. IEEE Radar Conf.*, Long Beach, CA, USA, Apr. 2002, pp. 311–317.
- [11] W. L. Melvin, M. J. Callahan, and M. C. Wicks, "Adaptive clutter cancellation in bistatic radar," in *Proc. 34th Asilomar Conf. Signals, Syst. Comput.*, Pacific Grove, CA, USA, Oct. 2000, pp. 1125–1130.
- [12] A. Fallah and H. Bakhshi, "Extension of adaptive angle-Doppler compensation (AADC) in STAP to increase homogeneity of data in airborne bistatic radar," in *Proc. 6th Int. Symp. Telecommun. (IST)*, Tehran, Iran, Nov. 2012, pp. 367–372.
- [13] V. Varadarajan and J. L. Krolik, "Joint space-time interpolation for distorted linear and bistatic array geometries," *IEEE Trans. Signal Process.*, vol. 54, no. 3, pp. 848–860, Mar. 2006.
- [14] B. Friedlander, "The MVDR beamformer for circular arrays," in *Proc. 34th Asilomar Conf. Signals, Syst. Comput.*, Pacific Grove, CA, USA, 2000, pp. 25–29.
- [15] M. Zatman, "The properties of adaptive algorithms with time varying weights," in *Proc. IEEE Sensor Array Multichannel Signal Process. Workshop*, Cambridge, MA, USA, Mar. 2000, pp. 82–86.
- [16] R. C. DiPietro, "Extended factored space-time processing for airborne radar systems," in *Proc. 26th Asilomar Conf. Signals, Syst., Comput.*, Pacific Grove, CA, USA, Oct. 1992, pp. 425–430.